Project RM 5

(Simulation of Interest Rate Dynamics)

Financial Engineering AY 2021-2022

Consider the European USD-denominated Swaption Market on May 26th, 2016. Students shall implement the Single-factor Hull-White Extended Vasicek (1994) model for the dynamics of interest rates [1 par 3.3]. The annex spreadsheet mktData_with_FRED.xlsx shows the quotes of interbank deposits (ACT/360); interest rate swaps (fixed leg annual bond basis, floating leg quarterly ACT/360) and the Black volatilities of ATM swaptions on the same IRS.

i) Implement the bootstrap of the zero-coupon curve from money market depos and IRS quotes (MID price) and calibrate the Single-factor Hull-White Extended Vasicek (1994) model ($a, \sigma \geq 0$) to the options with maturity 6 months and all underlying IRS maturities (1y, 2y, 3y, 4y, 5y): minimize the L2 price-distance. Derive from it and from the forward zero-coupon bonds' prices $B(t_0; t_0 + 0.50, t_0 + 1.50)$ the risk-neutral distribution six month later of the zero-coupon bonds' prices with one year maturity $B(t = 0.50; t, t + 1)$.

ii) You are an investor in fixed-income instruments. Derive from the results of the point i) the (risk-neutral) expected percentage return and its standard deviation of the following investment: buy on May 26th, 2016 a USD zero-coupon bond with maturity 18 months and sell it six months later.

iii) The Federal Reserve Bank of St. Louis (https://research.stlouisfed.org/fred2/) publishes the time series of the 6m Libor, 1y IRS, 2y IRS rate since year 2000 (attached spreadsheet, rates are in percentage). Students shall derive from it the time series of the zero-coupon curves (defined on three nodes: 6m, 1y, 2y) at semiannual frequency (last working day of the semester) from H1-2001 to H2-2015 and, based on these time series, students are required to evaluate the historical (real-word) distribution of realized percentage return of the investment into a zero-coupon bond with 18 months original maturity sold three months later.

iv) Students shall compare the expected return and its standard deviation of the investment according to the risk-neutral and real-world distributions. Do the two approaches lead to the same conclusions for the fixed-income investor?

v) Students shall repeat the calibration of point i) with input the ATM volatility of the options with maturity 3 years and all underlying IRS maturities (1y, 2y, 3y, 4y, 5y) and derive the risk-neutral distribution of the investment in the zero-coupon bond (same method as already done in point ii). Compare the results and discuss which one is the best calibration choice for the fixed-income investor defined above.

Realize a library in matlab.

[1] Brigo, D. and Mercurio, F. *Interest Rate Models - Theory and Practice*, 2001, Springer.

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